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*Feasibility Findings, Early Experiment, Design Direction,
Updated Risk Register, and AI Use Statement*

Team Carrillo · Individual enrollment · Option B, Instructor Assigned

13 September 2026

What this checkpoint is

This is the week the charter stops being a plan and starts being a test. The course packet asks whether the project I filed on 7 September is a project I can actually run on equipment I control, with data I actually have, before anyone spends the rest of the semester decorating a dead assumption. I treated it that way.

The chartered problem still stands: one attendant covering a bank of self-checkout kiosks; a freeze (will-not-scan, bag-scale mismatch, age-restricted item, unexpected item in bag) leaves a customer idle until that attendant walks over; other kiosks stop in the same window. The semester environment is still a mock. I do not have live lane logs, I do not have a store sponsor, and I am not putting work screenshots in a course repo.

1. Access confirmed in practice

The charter already ruled out live point-of-sale. Access for this project means three things I can touch this week: a machine that runs the mock, a labeled event file I authored, and a script that reads that file and writes an attendant log. Those three exist. A store feed does not, and I am not calling a missing feed “pending access.”

Need	Claimed in Week 1	Evidence this week	Verdict
Compute	Laptop or Colab	Local Python 3 with pandas and scikit-learn. Script ran to completion 13 Sept 2026. Elapsed fit for the throwaway model was 0.019 s. Rule evaluation on 20 injected freezes had a median latency under 1 ms per event on this machine.	Confirmed
Data	Staged events I create; no live customer records	File staged_events_sample.csv: 120 authored rows (60 freeze / 60 clear). Fields are mock-kiosk observables only: scan_ok, scale_delta_g, age_check_needed, unexpected_bag, idle_s, items_in_tx, attendant_busy. No names, phones, loyalty IDs, or store screenshots.	Confirmed — sample, not a store extract
Equipment / lane hardware	None. Mock only.	No kiosk sign-out. None requested. None required by the charter success tests.	Not needed; not faked
Credentials / sponsor login	Instructor as Option B owner	No third-party account. No HEB credential. Course work stays on a personal machine and will move to a public repo without customer data.	Confirmed as absence
Repository	Create after Week 1 record	Still not posted to Canvas as of 13 September. That is a process gap, not a data gap. The event file and script exist locally and will be the first commit.	Not yet public — see R7

Table 1. Access evidence. A downloaded sample counts. A promise from a store does not. I have the sample. I do not have the store.

Sample of the staged file (first six rows after shuffle)

event_id	scan_ok	scale_Ag	age	unexp.	idle_s	items	busy	kind	y
E014	0	11.2	0	0	7.81	9	1	wont_scan	1

event_id	scan_ok	scale_Δg	age	unexp.	idle_s	items	busy	kind	y
E067	1	6.4	0	0	1.22	4	0	none	0
E031	1	-186.0	0	0	5.44	12	1	bag_scale	1
E088	1	48.7	0	0	2.05	7	0	none	0
E022	1	9.8	1	0	13.10	3	1	age_restricted	1
E055	1	142.3	0	1	6.02	11	0	unexpected_bag	1

Table 2. Illustrative rows from the staged schema. The full 120-row file is submitted alongside this PDF as `staged_events_sample.csv`. Label `y` is held out of both candidate systems.

2. Riskiest assumption, and the throwaway test

If freeze versus clear is just a column I type, any “model” is a toy. The project dies if the only way to flag a freeze is to already know it is a freeze. The assumption that would kill the charter if false is therefore:

Kill assumption. Observable mock-kiosk fields that I can generate without a store system (scan result, bag-scale delta, age-check flag, unexpected-bag flag, idle seconds, basket size, whether the attendant is already busy) carry enough signal to flag freezes at something close to the Section 4 thresholds, without feeding the label back into the system.

That is the assumption I tested. Not “a grocery chain will give me an API.” That one is already false and already scoped out.

2.1 Method

I generated 120 staged events with a fixed seed (4376): 60 freezes split across the four floor types listed in the 8–13 September log, and 60 clears. Clears are not all clean. Some have heavy produce scale noise, a slow bagger, or an age flag that an attendant already cleared. Twelve percent of freezes are deliberately messy so the set is not a cartoon. An 80/40 stratified split held out 20 freeze and 20 clear events. The holdout is smaller than the charter’s 40+40 test. This week’s job was to see whether the assumption survives a first cut, not to claim the final score.

Two systems saw only the seven observable fields. Neither saw `freeze_kind` or `is_freeze` at decision time.

2.2 What happened

System	Holdout freeze hits	Holdout freeze misses	False flags on clears	Notes against charter
A. Hand rules	19 / 20	1	0 / 20	On this cut, hit rate is above the 34/40 spirit (85% would miss the bar; 95% would clear it). False flags under the 8/40 cap. Not yet the official 40+40 set.
B. Logistic regression	17 / 20	3	2 / 20	Trained on 80 rows. Weaker than the rules on the same holdout. Coefficients moved the right direction (scan_ok negative, idle_s and unexpected_bag positive) but the sample is too small to trust weights.
B-alt. Depth-3 tree (not selected)	20 / 20	0	2 / 20	Looks prettier. I do not trust a 20-hit tree on 80 homemade rows. Recorded only so I do not pretend the learned family had one member.

System	Holdout freeze hits	Holdout freeze misses	False flags on clears	Notes against charter
A, scored on all 120	59 / 60	1	2 / 60	Full-file check after the holdout, not a second independent test. One missed freeze, two false flags.

Table 3. Throwaway results, 13 September 2026. Charter thresholds remain 34/40 hits, ≤ 6 misses, ≤ 8 false flags on 40 clears, median notice ≤ 3 s.

Time-to-notice in the mock: I injected 20 staged freezes into a Python attendant log and measured machine time from inject to rule decision. Median was far under 3 seconds (sub-millisecond on this laptop). That test says the loop is not the bottleneck. It does not say a human looking at a screen will react in 3 seconds. Criterion 3 in the charter is still about the log appearing, not about gait speed.

2.3 Did the assumption fail?

It did not fail hard enough to kill the mock. Observable fields, with no label leakage, separated most of the staged freezes from most of the staged clears. It also did not pass in the way a store pilot would need. I wrote the data generator. I wrote the rules. The generator and the rules share a brain. That is the honest limit of this experiment, and it is why I am not printing “34/40 achieved.”

What I am doing about that: the next labeled set has to include cases I did not encode on purpose — events written down from memory after a shift, then coded by the seven fields only, with the freeze label applied last. If rules that looked good this week fall over on that set, the assumption gets retested and the register gets another line. Hiding the circularity until Week 12 would be the failure mode the prompt warned about.

3. Two candidate approaches

Both approaches have to live inside the charter: mock only, no live POS, no faces, demoable by someone else, four numbered tests, solo calendar, after-shift hours.

Approach A — Explicit rules plus an attendant log

If scan_ok is 0, flag. If $|\text{scale_delta}| \geq 75$ g, flag. If an age check is still open and idle time ≥ 5 s, flag. If unexpected_bag is 1, flag. If idle time ≥ 10 s with no other cause, flag. Write the hit to a timestamped log the attendant role can read. No training step. Every flag can be explained in one sentence.

Approach B — Learned classifier on the same fields

Fit a small model (logistic regression was the primary; a shallow tree was the check) on staged rows, emit a probability or a hard label, send the same attendant log. The pitch is that messy overlaps a person will not want to keep encoding as if-statements might still be separable.

Charter constraint	Approach A — rules + log	Approach B — learned model
No live store data	Fits. Rules read fields I invent or later log by hand.	Fits only if I keep generating or hand-coding rows. Starves without volume.
Someone else can rerun the tests	Yes. Rules are on one page. Thresholds are public.	Yes if the seed, file, and library versions ship. Harder to audit why a row flipped.
34/40 freeze / ≤ 8 false flags (spirit of the test)	Better on this week's holdout (19/20, 0 false flags on 20 clears).	Worse on the same cut (17/20, 2 false flags). Needs more honest rows, not more layers.
Median notice ≤ 3 s in the mock	Cleared by a wide margin on inject-to-log timing.	Also cleared. Inference is not the issue at this size.

Charter constraint	Approach A — rules + log	Approach B — learned model
Zero real customer identifiers	Holds if I never paste a receipt into the file.	Same rule. A model is not an excuse to add a camera.
Solo + shift hours (R3, R5)	Cheaper to maintain. I can change one threshold after a bad week without a retraining ritual.	Retrain cost is small in compute and large in discipline. Easy to hide behind “needs more data.”
Ethics: attendant judged on a bad log (charter §7)	A missed rule is visible. A supervisor can disagree with the threshold.	A silent weight is harder to argue with on a floor, even a mock floor.
Risk R4: pretty demo, ugly events	Ugly cases have to be written as rules or they are missed. That is visible.	Ugly cases can be absorbed into a score and look like success until the holdout changes.

Table 4. Fair comparison against the charter, not against a fantasy production stack.

4. Design direction

I am taking Approach A as the semester spine: explicit rules, staged event file, attendant log, the four tests from the charter. I am setting Approach B aside as the primary path, not as a banned idea.

Why A.

It won the only test I actually ran. It matches the operating environment I can defend in front of a reviewer (a log a person can read). It keeps Governance honest: if the system flags a customer-looking event, I can say which rule fired. It is the only version I can still finish if a work week eats the calendar, because there is no training loop to babysit.

Why not B as the lead.

The learned model did not beat the rules on a set I designed. On 80 homemade training rows, a better score next week would be as likely to mean I overfit the generator as it would mean I discovered a real pattern. A depth-3 tree that “hits 20/20” is exactly the pretty demo R4 warned about. I am not building the rest of the semester on that number.

B stays in the lab notebook as a foil. If a later hand-coded set from shift memory makes the rules collapse and a small model holds, I will reopen the decision in that week’s log and in the register. I will not reopen it because a library is fashionable.

5. Scope: confirmed, with one process cut and one test caveat

The charter scope is still achievable by Week 16 if “achievable” means the mock: a labeled event file, a working loop that takes a staged freeze and shows it on an attendant view, and the four tests run by a second person on held-out staged events. I am not revising the problem statement. I am not opening live POS. I am not adding theft prediction or a production app.

What changed this week is narrower. First, the official 40+40 holdout is not done; this checkpoint used 20+20 plus a full-file sanity pass. The charter thresholds stay. The date they get measured on the full set moves to the next build week, after the repo exists. Second, the GitHub URL promised after Week 1 is still missing. That does not shrink the technical scope. It does mean Systems and Integration is late, and I am writing that down instead of dating a blank link.

If something has to be cut later under R3, the cut order from the charter still holds: drop extra analysis before dropping the three numbered performance tests (hits, false flags, notice time).

6. Updated risk register

Week 1/M01 listed R1–R6. Feasibility work added three lines and changed likelihood on two existing ones. An unchanged register after this week would have meant the experiment was not done.

ID	Risk	Type	L	I	Change this week	Owner / move
R1	Instructor brief replaces the problem.	Scope	M	H	Unchanged. No new brief received as of 13 Sept.	Rewrite within one week if a packet lands.
R2	No real lane logs or hardware.	Access	H	M	Impact down from H to M. Charter already scoped the mock; this week proved the mock file and loop can exist.	Stay on staged events. Do not reopen store access as a dependency.
R3	Solo team. Shifts and illness slip the calendar.	Team	H	H	Unchanged. Optional sessions skipped. Hours this week were four on notes, plus this experiment.	Cut analysis before tests. Keep the weekly log.
R4	Clean staged data, ugly reality.	Technical	H	H	Likelihood up M→H. The generator and the rules share an author. That is now a measured problem, not a slogan.	Next set: label last, after coding fields from shift memory. Report misses.
R5	Too few weekday hours before a review.	Schedule	H	M	Unchanged.	Work after early-ending shifts. Say so in the log if a week is empty.
R6	Read as an official store project.	Ethics	M	H	Unchanged. This document repeats instructor-assigned mock.	No work screenshots. No store procedures in the repo.
R7	Repository still not in Canvas.	Process	H	M	New. Promised after Week 1. Still open 13 Sept.	Create the repo and commit csv + script before the next log.
R8	Circular generator/rules make Week 5 look better than Week 12.	Technical / integrity	H	H	New. Directly from the throwaway test.	Independent hand-coded set. Do not tune rules on the official holdout.
R9	Second-person scorer for the 40+40 test never appears.	Schedule / method	M	H	New. Charter requires a second person to score hits and misses. Team of one has to recruit a scorer or ask the instructor how solo scoring works.	Ask in writing. If no scorer, document the substitute method before collecting the official set.

Table 5. Register after feasibility. L = likelihood, I = impact. H high, M medium, L low.

7. Ethics reminder (unchanged, restated)

Who could be harmed if this were live: a customer left idle because of a miss; a customer hovered over because of a false flag; an attendant scored on a bad log. This semester those harms are contained because there is no live connection. That containment is a scope choice, not a claim that the rules are safe on a floor. I will not pull receipts, face video, or employee performance files from work to “improve” the mock.

8. AI use statement

Tool: Grok (xAI). What it was used for: turning this week’s notes and the course rubric into a PDF layout; drafting section order so the four graded criteria are impossible to miss; helping write and run the throwaway Python generator and scorers; tightening sentences. What it was not used for: inventing a store sponsor, inventing lane credentials, pasting internal store screens, or reporting success numbers from a system that was not run. The 19/20, 17/20, and 59/60 figures in Table 3 came from feasibility_experiment.py on 13 September 2026 against staged_events_sample.csv. I read every section before submitting. Thresholds in the M01 charter are still targets, not medals.

9. What I am turning in with this PDF

File	What it is
M05_Carrillo_Carrillo_ITAI4376.pdf	This findings document.
staged_events_sample.csv	120-row mock event file. Access evidence.
holdout_predictions.csv	40-row holdout with rule and logistic predictions.
experiment_summary.json	Machine-readable metrics from the throwaway run.
feasibility_experiment.py	Script that built the file and scored both approaches. Re-runnable with seed 4376.

Table 6. Evidence packet. The PDF is the narrative. The csv and script are the access proof.

10. Bottom line

I can do the project I chartered, as a mock, on a machine I own, with data I authored. I cannot do it as a store system, and I am not waiting for that door to open. The riskiest assumption — that mock observables carry freeze signal without label leakage — survived a first, biased test and is now the thing I have to test more honestly. Rules plus a log is the direction. The learned model is on the bench. Scope holds. The register grew because the work did.

Signature: Elizabeth Carrillo

Date: 13 September 2026

Team: Carrillo (individual) · Option B · Instructor assigned mock, not a store pilot.